

A Practical Guide to Image to image translation

TOPIC -- IMAGE TO IMAGE TRANSLATION

-- 01 --

Parametric methods Lagrangian techniques are based on parameterizing the contour according to some sampling strategy and then evolving each element according to image and internal terms. Such techniques are fast and efficient, however the original "purely parametric" formulation (due to Kass, Witkin and Terzopoulos in 1987 and known as "snakes"), is generally criticized for its limitations regarding the choice of sampling strategy, the internal geometric properties of the curve, topology changes (curve splitting and merging), addressing problems in higher dimensions, etc.. Nowadays, efficient "discretized" formulations have been developed to address these limitations while maintaining high efficiency. In both cases, energy minimization is generally conducted using a steepest-gradient descent, whereby derivatives are computed using, e.g., finite differences.

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$$\alpha (1 - \delta (\| \mathbf{i} - \mathbf{initial} \|)) + \beta \sum_{q \in N(i)} (1 - \delta (\| \mathbf{i} - \mathbf{q}(i) \|)) . \quad \{\displaystyle \alpha (1 - \delta (\| \mathbf{i} - \mathbf{initial} \|)) + \beta \sum_{q \in N(i)} (1 - \delta (\| \mathbf{i} - \mathbf{q}(i) \|)) .\}$$

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Related images such as a photo album or a sequence of video frames often contain semantically similar objects and scenes, therefore it is often beneficial to exploit such correlations. The task of simultaneously segmenting scenes from related images or video frames is termed co-segmentation, which is typically used in human action localization. Unlike conventional bounding box-based object detection, human action localization methods provide finer-grained results, typically per-image segmentation masks delineating the human object of interest and its action category (e.g., Segment-Tube). Techniques such as dynamic Markov Networks, CNN and LSTM are often employed to exploit the inter-frame correlations.

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Groups of image segmentation Semantic segmentation is an approach detecting, for every pixel, the belonging class. For example, in a figure with many people, all the pixels belonging to persons will have the same class id and the

pixels in the background will be classified as background. Instance segmentation is an approach that identifies, for every pixel, the specific belonging instance of the object. It detects each distinct object of interest in the image. For example, when each person in a figure is segmented as an individual object. Panoptic segmentation combines both semantic and instance segmentation. Like semantic segmentation, panoptic segmentation is an approach that identifies, for every pixel, the belonging class. Moreover, like in instance segmentation, panoptic segmentation distinguishes different instances of the same class.

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Optimization algorithms Each optimization algorithm is an adaptation of models from a variety of fields and they are set apart by their unique cost functions. The common trait of cost functions is to penalize change in pixel value as well as difference in pixel label when compared to labels of neighboring pixels.

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Region-growing methods Region-growing methods rely mainly on the assumption that the neighboring pixels within one region have similar values. The common procedure is to compare one pixel with its neighbors. If a similarity criterion is satisfied, the pixel can be set to belong to the same cluster as one or more of its neighbors. The selection of the similarity criterion is significant and the results are influenced by noise in all instances. The method of Statistical Region Merging (SRM) starts by building the graph of pixels using 4-connectedness with edges weighted by the absolute value of the intensity difference. Initially each pixel forms a single pixel region. SRM then sorts those edges in a priority queue and decides whether or not to merge the current regions belonging to the edge pixels using a statistical predicate. One region-growing method is the seeded region growing method. This method takes a set of seeds as input along with the image. The seeds mark each of the objects to be segmented. The regions are iteratively grown by comparison of all unallocated neighboring pixels to the regions. The difference between a pixel's intensity value and the region's mean, δ , is used as a measure of similarity. The pixel with the smallest difference measured in this way is assigned to the respective region. This process continues until all pixels are assigned to a region. Because seeded region growing requires seeds as additional input, the segmentation results are dependent on the choice of seeds, and noise in the image can cause the seeds to be poorly placed. Another region-growing method is the unseeded region growing method. It is a modified algorithm that does not require explicit seeds. It starts with a single region A_1 —the pixel chosen here does not markedly influence the

final segmentation. At each iteration it considers the neighboring pixels in the same way as seeded region growing. It differs from seeded region growing in that if the minimum δ is less than a predefined threshold T then it is added to the respective region A_j . If not, then the pixel is considered different from all current regions A_i and a new region A_{n+1} is created with this pixel. One variant of this technique, proposed by Haralick and Shapiro (1985), is based on pixel intensities. The mean and scatter of the region and the intensity of the candidate pixel are used to compute a test statistic. If the test statistic is sufficiently small, the pixel is added to the region, and the region's mean and scatter are recomputed. Otherwise, the pixel is rejected, and is used to form a new region. A special region-growing method is called λ -connected segmentation (see also λ -connectedness). It is based on pixel intensities and neighborhood-linking paths. A degree of connectivity (connectedness) is calculated based on a path that is formed by pixels. For a certain value of λ , two pixels are called λ -connected if there is a path linking those two pixels and the connectedness of this path is at least λ . λ -connectedness is an equivalence relation. Split-and-merge segmentation is based on a quadtree partition of an image. It is sometimes called quadtree segmentation. This method starts at the root of the tree that represents the whole image. If it is found non-uniform (not homogeneous), then it is split into four child squares (the splitting process), and so on. If, in contrast, four child squares are homogeneous, they are merged as several connected components (the merging process). The node in the tree is a segmented node. This process continues recursively until no further splits or merges are possible. When a special data structure is involved in the implementation of the algorithm of the method, its time complexity can reach $O(n \log n)$, an optimal algorithm of the method.